SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	D.G.RAGHAVI	**Reg. No.:**	192521173
Date:	03/08/2026	**Duration:**	60 minutes

Code of Conduct

I D.G.RAGHAVI(192521173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ______________________

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario** ○ Carefully understand the given problem statement.
    - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements** ○ List all functional requirements required by the proposed system.
    - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. **Identify Actors and Use Cases** ○ Identify all primary and secondary actors interacting with the system. ○ List the major use cases associated with each actor.
    - ○ Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)** Develop an SRS document containing: ○ Introduction ○ Problem Description
    - ○ Scope of the System ○ Functional Requirements ○ Non-Functional Requirements ○ Use Case Descriptions ○ Data Requirements ○ External Interface Requirements ○ System Features
5. **Identify Assumptions and Constraints** ○ List the assumptions considered while developing the system.
    - ○ Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency** ○ Verify whether all stakeholder requirements have been addressed. ○ Check for ambiguity, redundancy, inconsistency, and missing requirements.
    - ○ Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.

Software Requirement Specification	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or
(SRS) Documentation	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

NAME :D.G.RAGHAVI **COURSE CODE** : CSA1013
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CO2 - ASSESSMENT TOOL – 01

Q27. AI-Powered Personalized Nutrition Recommendation System

1.1 Business Domain

The proposed system belongs to the Construction Management, Civil Engineering, Internet of Things (IoT), Computer Vision, and Artificial Intelligence (AI) domain. It is designed to improve the efficiency of construction project management by providing real-time monitoring and intelligent decision-making capabilities. Construction projects involve multiple activities such as material management, labor supervision, equipment utilization, safety inspections, and project scheduling. Traditional monitoring methods mainly depend on manual site inspections and periodic reports, which often result in delayed identification of problems and inaccurate project tracking.

The proposed AI-powered construction monitoring platform integrates advanced technologies such as drones, surveillance cameras, IoT sensors, cloud computing, and Artificial Intelligence to continuously collect and analyze construction site data. The system automatically monitors project progress, identifies safety violations, detects construction risks, predicts schedule delays, and provides real-time insights to project managers. By replacing manual supervision with intelligent automation, the platform improves project efficiency, enhances worker safety, reduces operational costs, and ensures timely project completion.

1.2 Stakeholders

The proposed construction monitoring system involves several stakeholders who contribute to the successful execution and management of construction projects. The primary stakeholder is the Project Manager, who is responsible for planning, monitoring, and controlling construction activities. The system enables project managers to monitor real-time project progress, allocate resources efficiently, identify potential delays, and make informed decisions based on AI-generated insights.

The Site Engineer is another important stakeholder who supervises day-to-day construction activities. Site engineers use the platform to monitor ongoing work, verify project milestones, inspect structural progress, and ensure that construction activities comply with project specifications. The real-time monitoring system helps them quickly identify issues and take corrective actions.

The Construction Workers indirectly benefit from the system because AI continuously monitors the construction site for safety compliance. The platform detects unsafe behavior such as the absence of helmets or safety jackets, entry into restricted areas, and unsafe working practices. Early identification of safety violations helps reduce workplace accidents and ensures a safer working environment.

The Safety Officer uses the platform to monitor safety standards across the construction site. The AI system automatically detects hazards, equipment misuse, unsafe working conditions, and potential accident-prone areas. Safety officers can immediately respond to alerts and implement preventive measures to minimize workplace risks.

The Company Management or Construction Firm benefits by obtaining real-time project analytics, resource utilization reports, financial performance indicators, and project completion forecasts. This helps improve productivity, reduce project costs, and increase customer satisfaction.

The System Administrator manages the technical aspects of the platform, including user management, database maintenance, AI model updates, cloud infrastructure, system security, software maintenance, and backup operations. The administrator ensures continuous system availability and secure storage of construction data.

1.3 Existing Challenges

The construction industry faces several challenges that affect project completion, cost efficiency, and worker safety. One of the major challenges is the lack of real-time project visibility. Most construction companies rely on manual inspections and weekly or monthly progress reports, making it difficult to identify issues immediately. As a result, project delays often remain unnoticed until they become significant.

Another major challenge is inefficient resource utilization. Construction resources such as labor, machinery, equipment, and materials are often not utilized optimally because managers lack continuous monitoring systems. Poor resource allocation increases project costs, reduces productivity, and delays project completion.

Safety management is another critical concern in construction projects. Construction sites are highly hazardous environments where workers are exposed to risks such as falling objects, heavy machinery, electrical hazards, and unsafe working conditions. Manual safety inspections cannot continuously monitor every activity, resulting in delayed identification of safety violations and increased accident risks.

Project schedule deviations are also common because unforeseen events such as equipment failures, weather conditions, labor shortages, or delayed material delivery are not detected early. Traditional project management methods cannot accurately predict future delays, making proactive planning difficult.

Risk assessment is another limitation of conventional construction monitoring systems. Manual risk evaluation depends heavily on human observation and experience, which may overlook hidden risks. Without predictive analytics, project managers often respond to problems only after they have occurred instead of preventing them beforehand.

Additionally, maintaining accurate documentation and communication among project stakeholders is challenging. Delays in information sharing often lead to poor decision-making, reduced coordination, and increased project costs. These challenges emphasize the need for an intelligent AI-powered construction monitoring platform capable of providing continuous monitoring, predictive analytics, automated safety inspections, and real-time decision support.

1.4 System Objectives

The primary objective of the proposed AI-Powered Construction Monitoring Platform is to develop an intelligent system that continuously monitors construction activities using Artificial Intelligence, drones, surveillance cameras, and IoT sensors. The system aims to improve project execution by providing real-time visibility into construction progress and enabling project managers to make informed decisions.

Another important objective is to automatically track construction progress by analyzing live images and sensor data collected from the construction site. The system compares actual project progress with planned schedules and identifies deviations at an early stage. This enables project managers to implement corrective measures before delays become critical.

The system also aims to improve worker safety by automatically detecting safety violations such as missing protective equipment, unauthorized access to hazardous areas, unsafe worker behavior, and dangerous working conditions. Immediate alerts allow safety officers to respond quickly and reduce workplace accidents.

Efficient resource utilization is another key objective of the proposed platform. By monitoring labor productivity, equipment usage, machinery performance, and material availability, the system helps construction companies optimize resource allocation and reduce operational costs.

The platform further aims to assess project risks using Artificial Intelligence and predictive analytics. The AI engine analyzes historical project data, current site conditions, environmental factors, and resource utilization patterns to identify potential risks and predict schedule deviations before they occur. This allows proactive risk management and improves project planning.

Another objective is to generate comprehensive dashboards and analytical reports that provide project managers with real-time insights into construction progress, safety compliance, resource utilization, and project performance. These reports support strategic planning and improve communication among project stakeholders.

Finally, the proposed system aims to improve overall project efficiency, reduce construction delays, minimize safety hazards, optimize costs, enhance decision-making, and ensure successful project completion through intelligent monitoring and predictive analytics.

2. Functional Requirements

Functional requirements describe the services and operations that the AI-Powered Construction Monitoring Platform must perform to meet the needs of construction companies, project managers, engineers, and other stakeholders. These requirements define how the system collects real-time site information, processes construction data, analyzes project progress, detects safety violations, predicts risks, and supports decision-making throughout the construction lifecycle.

2.1 User Management

The User Management module is responsible for providing secure access to the construction monitoring platform. The system shall allow different categories of users, including project managers, site engineers, safety officers, construction supervisors, and system administrators, to register and access the application using authenticated login credentials. During registration, users must provide personal and professional information such as name, employee ID, email address, mobile number, designation, and organization details. The system should verify the information before creating the user account and securely store login credentials using password encryption techniques. Users should also have the ability to reset forgotten passwords through email or mobile verification. Depending on their role, the system grants appropriate access permissions so that each user can only view or modify authorized information. The user management module also enables administrators to create, modify, disable, or delete user accounts while maintaining complete audit logs of user activities.

2.2 Construction Site Data Management

The Construction Site Data Management module is responsible for collecting, storing, and organizing all information related to the construction site. The system continuously gathers data from multiple sources such as drones, CCTV cameras, IoT sensors, GPS devices, environmental monitoring sensors, and manually entered project records. Information including project details, construction phases, equipment status, worker attendance, machinery utilization, weather conditions, material inventory, and structural progress is securely stored in the centralized database. The module validates incoming data to eliminate inconsistencies and duplicates before processing. Historical site information is also maintained to support project analysis, reporting, and predictive modeling. By maintaining a centralized repository of construction data, the system enables efficient monitoring and informed decision-making throughout the project lifecycle.

2.3 AI-Based Project Progress Monitoring

The AI-Based Project Progress Monitoring module continuously evaluates construction progress using images captured by drones and surveillance cameras along with data collected from IoT sensors. Artificial Intelligence algorithms compare the actual construction status with the planned project schedule to determine whether the project is progressing as expected. The system automatically identifies completed activities, ongoing work, and delayed tasks by analyzing visual information and construction milestones. Progress percentages are calculated for different sections of the project and presented through graphical dashboards. Whenever significant deviations from the planned schedule are detected, the system immediately notifies project managers so that corrective actions can be initiated. This automated monitoring significantly reduces the need for manual inspections and improves the accuracy of project tracking.

2.4 Safety Monitoring and Violation Detection

The Safety Monitoring module plays a critical role in improving worker safety on construction sites. Using computer vision techniques and Artificial Intelligence, the system continuously analyzes live video streams from surveillance cameras and drone footage to identify unsafe working conditions and safety violations. The system detects whether workers are wearing mandatory personal protective equipment such as helmets, safety jackets, gloves, and safety shoes. It also identifies unauthorized entry into restricted areas, improper equipment usage, unsafe worker behavior, fire hazards, and structural risks. Whenever a safety violation is detected, the system immediately generates alerts and sends notifications to safety officers and project managers. Incident reports are automatically created and stored for future analysis. This functionality helps reduce workplace accidents, ensures compliance with safety regulations, and creates a safer construction environment.

2.5 Project Risk Assessment and Prediction

The Project Risk Assessment module utilizes Artificial Intelligence and predictive analytics to evaluate potential risks that may affect project completion. The AI engine analyzes historical project data, current construction progress, labor productivity, equipment utilization, environmental conditions, and resource availability to identify possible risk factors. The system predicts schedule deviations, cost overruns, equipment failures, material shortages, and productivity losses before they occur. Risk scores are generated for different project components, allowing managers to prioritize corrective actions. The platform also recommends preventive measures to minimize identified risks and improve overall project performance. Predictive risk assessment enables proactive project management and reduces unexpected disruptions during construction.

2.6 Resource Monitoring and Management

Efficient resource utilization is essential for successful construction project execution. The Resource Monitoring module continuously tracks the usage of labor, construction equipment, machinery, materials, and vehicles throughout the project. The system monitors equipment operating hours, worker productivity, material consumption, fuel usage, and machinery performance using IoT sensors and management records. AI algorithms analyze utilization patterns to identify idle resources, equipment overloading, or material wastage. Resource allocation recommendations are generated to improve operational efficiency and reduce project costs. Managers can also view resource availability through interactive dashboards and optimize workforce deployment based on project requirements.

2.7 Notifications and Alert Management

The Notification and Alert module ensures that all project stakeholders receive timely updates regarding project activities and critical events. The system automatically generates notifications whenever project delays, safety violations, equipment failures, abnormal sensor readings, or risk predictions are detected. Alerts are delivered through mobile applications, email, SMS, or dashboard notifications depending on user preferences. Daily progress updates, milestone completion reminders, inspection schedules, maintenance alerts, and emergency notifications are also provided. These real-time alerts enable project managers and safety officers to respond immediately to issues, thereby improving decision-making and minimizing project disruptions.

2.8 Report Generation and Dashboard Management

The Reporting module provides comprehensive analytical reports and visual dashboards that summarize construction activities and project performance. The system automatically generates daily, weekly, and monthly reports containing project progress, resource utilization, safety incidents, equipment performance, project risks, and schedule deviations. Interactive dashboards display charts, graphs, progress indicators, heat maps, and performance metrics that enable project managers to quickly understand the overall status of construction activities. Reports can be exported in various formats such as PDF and Excel for documentation and presentation purposes. These reports support strategic planning, project evaluation, and management decision-making.

2.9 Administrator Management

The Administrator module provides complete control over the operation and maintenance of the AI-powered construction monitoring platform. Administrators are responsible for managing user accounts, assigning user roles, updating project information, maintaining AI models, configuring IoT devices, managing drone and camera integrations, monitoring server performance, and maintaining the centralized database. The administrator also performs system backups, software updates, security monitoring, and troubleshooting to ensure uninterrupted system operation. Audit logs are maintained to monitor user activities and improve system accountability. Effective administration ensures the reliability, security, and continuous availability of the construction monitoring platform.

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3. Non-Functional Requirements

Non-functional requirements define the quality attributes that ensure the system performs efficiently, securely, and reliably while meeting user expectations. These requirements specify the performance standards, security measures, usability characteristics, scalability, maintainability, reliability, and availability of the AI-powered construction monitoring platform.

3.1 Performance

The system should provide high-performance processing by continuously collecting and analyzing data from drones, surveillance cameras, and IoT sensors in real time. Project progress analysis, safety violation detection, and risk prediction should be completed within a few seconds after receiving new data. The platform must support simultaneous monitoring of multiple construction sites and accommodate a large number of concurrent users without reducing system performance. Efficient cloud computing resources and optimized AI algorithms should be utilized to ensure smooth operation even during peak workloads.

3.2 Security

Security is a critical requirement because the system stores confidential construction project information, employee records, financial data, and surveillance footage. The platform must implement secure user authentication, encrypted password storage, HTTPS communication, and role-based access control to prevent unauthorized access. All construction data should be encrypted during transmission and storage. Regular security audits, firewall protection, intrusion detection mechanisms, and automatic backups should be implemented to protect the system against cyber threats and data breaches. User activity logs should also be maintained for accountability and security monitoring.

3.3 Reliability

The AI-powered construction monitoring platform should provide reliable operation with at least 99.9% system availability. The system should continue functioning even during temporary network interruptions or hardware failures by utilizing cloud redundancy and automatic failover mechanisms. Backup servers and disaster recovery procedures should ensure that project data is never permanently lost. The system should also detect software errors automatically and recover normal operation with minimal downtime.

3.4 Usability

The application should provide a simple, intuitive, and user-friendly interface that enables construction professionals to easily navigate the system without requiring extensive technical training. Dashboards should present project progress, safety alerts, and risk indicators using clear charts, graphs, and visual representations. Mobile compatibility should allow engineers and project managers to monitor construction activities from remote locations using smartphones and tablets. The interface should minimize user effort while maximizing operational efficiency.

3.5 Scalability

The system should be designed using scalable cloud architecture capable of supporting increasing numbers of construction projects, users, IoT devices, drones, surveillance cameras, and sensor networks. As construction companies expand their operations, the platform should accommodate additional monitoring sites without requiring major architectural modifications. The modular software design should also support future integration with Building Information Modeling (BIM), Geographic Information Systems (GIS), autonomous drones, and advanced AI technologies.

3.6 Maintainability

The software should be developed using modular programming techniques that simplify future maintenance activities. Developers should be able to update AI algorithms, modify monitoring rules, improve dashboards, fix software bugs, and upgrade system components without affecting other modules. Proper documentation, coding standards, version control, and testing procedures should be maintained to ensure long-term software quality and facilitate continuous improvement.

3.7 Availability

The AI-powered construction monitoring platform should provide uninterrupted 24×7 availability to enable continuous monitoring of construction activities throughout the project duration. Cloud deployment should ensure that authorized users can access the system from any location with an internet connection. Automatic server monitoring, load balancing, backup systems, and disaster recovery mechanisms should guarantee continuous access to project information even during unexpected failures or maintenance activities. Continuous system availability ensures effective project supervision, timely decision-making, and improved overall construction management.

4.1 Introduction

The AI-Powered Construction Monitoring Platform is an intelligent software solution developed to improve the efficiency, safety, and quality of construction project management. Construction projects involve multiple stakeholders, complex workflows, expensive resources, and strict deadlines, making continuous monitoring essential for successful project execution. Traditional monitoring methods primarily rely on manual inspections, paper-based documentation, and periodic progress reports, which often fail to provide real-time visibility into project activities. These limitations can result in project delays, resource wastage, safety hazards, and increased operational costs.

The proposed system integrates Artificial Intelligence (AI), drones, surveillance cameras, IoT sensors, cloud computing, and data analytics to provide real-time monitoring of construction activities. The platform continuously collects and analyzes data from various sources to evaluate project progress, monitor worker safety, detect construction risks, and predict schedule deviations. By automating construction monitoring and providing intelligent insights, the system supports project managers, engineers, and safety officers in making informed decisions. Ultimately, the platform aims to improve project execution, enhance workplace safety, optimize resource utilization, reduce delays, and ensure successful completion of construction projects.

4.2 Problem Description

Construction projects often experience delays, cost overruns, and quality issues because project monitoring depends heavily on manual inspections and periodic reporting. Manual supervision cannot continuously observe every activity occurring at large construction sites, making it difficult to identify problems at an early stage. As a result, project managers frequently become aware of schedule deviations, safety violations, equipment failures, or resource shortages only after they have significantly affected project performance.

Safety management is another major concern in the construction industry. Construction workers operate in hazardous environments where accidents may occur due to the absence of personal protective equipment, unsafe working practices, heavy machinery operation, or structural instability. Traditional safety inspections are performed only at specific intervals and cannot provide continuous monitoring of worker behavior or hazardous conditions.

Resource management also presents significant challenges because labor, machinery, and construction materials are often not utilized efficiently. Without real-time monitoring, idle equipment, material wastage, and low workforce productivity increase project costs and reduce overall efficiency. Furthermore, project managers face difficulties in predicting future delays and identifying potential risks because conventional monitoring systems lack intelligent analytical capabilities.

The proposed AI-powered construction monitoring platform addresses these limitations by providing continuous project monitoring, automated safety inspections, predictive risk analysis, intelligent progress tracking, and real-time decision support. The system enables proactive project management by identifying issues before they become critical, thereby improving construction quality, safety, productivity, and overall project performance.

4.3 Scope of the System

The scope of the proposed system covers the complete monitoring and management of construction projects from project initiation to completion. The platform continuously collects construction site information using drones, CCTV cameras, IoT sensors, GPS devices, and manual project records. It analyzes construction progress, monitors equipment performance, tracks worker activities, evaluates environmental conditions, and generates real-time insights using Artificial Intelligence.

The system supports project managers by comparing actual construction progress with planned schedules and identifying project delays at an early stage. Safety officers receive automated alerts whenever unsafe working conditions or safety violations are detected. Engineers can monitor structural development, resource utilization, and project milestones through interactive dashboards.

The platform also performs predictive analysis to identify potential project risks, schedule deviations, and resource shortages before they occur. Comprehensive analytical reports and dashboards provide management with valuable information for strategic planning and decision-making. The system is designed for construction companies, infrastructure projects, commercial buildings, industrial facilities, residential developments, and public works projects. Future enhancements may include Building Information Modeling (BIM) integration, autonomous drone inspections, and digital twin technology.

4.4 Functional Requirements

The AI-powered construction monitoring platform provides several functional capabilities that support intelligent project management and monitoring. The system allows authorized users such as project managers, engineers, safety officers, and administrators to securely register, log in, and access project information according to their assigned roles. Construction site data collected from drones, surveillance cameras, IoT sensors, and manual entries are securely stored and processed by the system.

The Artificial Intelligence engine continuously analyzes visual data and sensor readings to determine construction progress and compare actual work with planned project schedules. Whenever project delays or deviations are detected, the system immediately generates alerts and recommends corrective actions. AI algorithms also identify safety violations by detecting workers who fail to wear protective equipment, enter restricted areas, or perform unsafe activities.

The platform continuously monitors construction resources such as labor, machinery, equipment, and building materials to improve resource utilization and minimize wastage. Predictive analytics evaluate historical project information and current construction conditions to identify potential risks, equipment failures, and schedule deviations before they occur.

The system automatically generates graphical dashboards, analytical reports, project performance summaries, safety reports, and resource utilization reports. Notifications and alerts are delivered through mobile applications, email, SMS, and web dashboards to ensure that project stakeholders receive timely updates. Administrators maintain user accounts, monitor AI system performance, update project databases, configure IoT devices, and ensure continuous system operation.

4.5 Non-Functional Requirements

The AI-powered construction monitoring platform must satisfy several quality requirements to ensure efficient, secure, and reliable operation. The system should provide high performance by processing construction site data and generating AI-based analysis within a few seconds. It must support multiple construction sites and thousands of concurrent users without affecting response time or system stability.

Security is one of the most important non-functional requirements because the system stores confidential project information, surveillance footage, financial records, and employee details. Strong authentication, encrypted communication, secure cloud storage, role-based authorization, and regular security audits must be implemented to protect sensitive information from unauthorized access.

The system should maintain high reliability with continuous monitoring, automatic backup, disaster recovery, and fault-tolerant cloud infrastructure to ensure uninterrupted operation. User interfaces should be simple, intuitive, and compatible with desktop computers, tablets, and mobile devices, allowing construction professionals to access project information from any location.

The platform should also be scalable so that additional construction sites, drones, sensors, AI models, and monitoring devices can be integrated without significant architectural changes. Modular software architecture should simplify maintenance, software updates, AI model improvements, and future feature enhancements. Cloud deployment should ensure twenty-four-hour availability and enable remote access for authorized users.

4.6 Use Case Descriptions

Use Case 1: User Registration

Actor: Project Manager, Engineer, Safety Officer, Administrator

Description:
The registration process allows authorized construction personnel to create user accounts within the system. The user enters personal information, employee identification details, organization name, email address, mobile number, designation, and password. The system verifies the information, encrypts the password, assigns an appropriate user role, and stores the account securely in the database. After successful registration, users can access the system according to their assigned permissions.

Precondition:
The user must be an authorized employee of the construction organization.

Postcondition:
A secure user account is successfully created and stored in the system.

Use Case 2: Monitor Construction Progress

Actor: Project Manager, Site Engineer

Description:
The system continuously collects images from drones, surveillance cameras, and IoT sensor data from the construction site. Artificial Intelligence analyzes the collected information, identifies completed construction activities, calculates project completion percentages, compares progress with planned schedules, and generates visual progress reports. Whenever delays or deviations are detected, the system immediately notifies project managers for corrective action.

Precondition:
Construction monitoring devices must be connected and operational.

Postcondition:
Real-time project progress information and analytical reports are generated.

Use Case 3: Detect Safety Violations

Actor: Safety Officer

Description:
The AI engine continuously monitors construction workers using computer vision techniques. It detects whether workers wear helmets, safety jackets, gloves, and other protective equipment. The system also identifies unsafe behaviors, hazardous working conditions, and unauthorized access to restricted areas. Whenever violations are detected, instant alerts are sent to safety officers, and incident reports are automatically generated for future reference.

Precondition:
Surveillance cameras must continuously monitor construction activities.

Postcondition:
Safety violations are recorded, and emergency notifications are generated.

Use Case 4: Predict Project Risks

Actor: Project Manager

Description:
The Artificial Intelligence engine evaluates project schedules, historical construction records, equipment utilization, labor productivity, weather conditions, and material availability. Using predictive analytics, the system identifies potential risks, predicts schedule deviations, estimates project completion dates, and recommends preventive actions that minimize project delays and cost overruns.

Precondition:
Historical and current project data must be available.

Postcondition:
Risk reports and predictive recommendations are generated.

4.7 Data Requirements

The system requires accurate and continuously updated information for effective construction monitoring. Project data includes project identification numbers, project schedules, construction milestones, contractor information, work packages, and completion status. Employee records contain worker information, engineer details, attendance records, safety certifications, and assigned responsibilities.

Construction monitoring data includes drone images, surveillance videos, IoT sensor readings, GPS coordinates, equipment operating information, environmental conditions, weather data, and machinery performance records. Safety information consists of worker protective equipment status, accident reports, safety inspection reports, hazard records, and emergency alerts. Analytical data generated by the AI engine includes project completion percentages, productivity indicators, equipment utilization, risk assessments, delay predictions, and performance metrics. All information is securely stored within centralized cloud databases for analysis, reporting, and future reference.

4.8 External Interface Requirements

The proposed system provides a user-friendly web application and mobile application that enable construction professionals to monitor project activities from any location. Interactive dashboards display project progress, resource utilization, safety alerts, and AI-generated recommendations through graphical visualizations.

The hardware interface includes drones, CCTV cameras, IoT sensors, GPS tracking devices, environmental monitoring equipment, cloud servers, smartphones, tablets, desktop computers, and laptops. The software interface integrates Artificial Intelligence algorithms, cloud databases, computer vision models, IoT communication platforms, GIS services, and project management software.

The communication interface utilizes secure internet connectivity, HTTPS protocols, wireless sensor networks, Wi-Fi, cellular networks, email services, SMS gateways, and cloud communication services to enable secure data transmission between construction sites and the monitoring platform.

4.9 System Features

The AI-Powered Construction Monitoring Platform provides several advanced features that significantly improve project management. These include real-time construction monitoring using drones and IoT sensors, AI-based project progress tracking, automated safety violation detection, predictive project risk analysis, intelligent resource monitoring, equipment performance analysis, schedule deviation prediction, automated alerts, cloud-based dashboards, graphical project reports, and mobile accessibility.

The system also supports role-based user management, centralized project databases, automated report generation, remote project supervision, predictive maintenance recommendations, and continuous AI-driven decision support. By combining Artificial Intelligence, IoT technology, computer vision, and cloud computing, the platform enables construction companies to improve productivity, reduce project delays, enhance worker safety, optimize resource utilization, and successfully complete projects within budget and schedule.